THE VIRIAL THEOREM IN STELLAR ASTROPHYSICS

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CHAPTER OVERVIEW

1: Development of the Virial Theorem

1.1: The Basic Equations of Structure

Before turning to the derivation of the virial theorem, it is appropriate to review the origin of the fundamental structural Equations of stellar astrophysics. This not only provides insight into the basic conservation laws implicitly assumed in the description of physical systems, but by their generality and completeness graphically illustrates the complexity of the complete description that we seek to circumvent. Since lengthy and excellent texts already exist on this subject, our review will of necessity be a sketch. Any description of a physical system begins either implicitly or explicitly from certain general conservation principles. Such a system is considered to be a collection of articles, each endowed with a spatial location and momentum which move under the influence of known forces. If one regards the characteristics of spatial position and momentum as being highly independent, then one can construct a multi-dimensional space through which the particles will trace out unique paths describing their history.

This is essentially a statement of determinism, and in classical terms is formulated in a six-dimensional space called phase-space consisting of three spatial dimensions and three linearly independent momentum dimensions. If one considers an infinitesimal volume of this space, he may formulate a very general conservation law which simply says that the divergence of the flow of particles in that volume is equal to the number created or destroyed within that volume.

The mathematical formulation of this concept is usually called the Boltzmann transport Equation and takes this form:

$$\frac{\partial \psi}{\partial \mathrm{t}}+\sum_{\mathrm{i}=1}^{3} \dot{\mathrm{x}}_{\mathrm{i}} \frac{\partial \psi}{\partial \mathrm{x}_{\mathrm{i}}}+\sum_{\mathrm{i}=1}^{3} \dot{\mathrm{p}}_{\mathrm{i}} \frac{\partial \psi}{\partial \mathrm{p}_{\mathrm{i}}}=\mathrm{S},$$

or in vector notation

$$\frac{\partial \psi}{\partial \mathrm{t}}+\mathbf{v} \cdot \nabla \psi+\mathbf{f} \cdot \nabla_{\mathrm{p}} \psi=\mathrm{S}, \tag{1.1.1}$$

where ψ is the density of points in phase space, f is the vector sum of the forces acting on the particles and S is the 'creation rate' of particles within the volume. The homogeneous form of this Equation is often called the Louisville Theorem and would be discussed in detail in any good book on Classical Mechanics.

A determination of ψ as a function of the coordinates and time constitutes a complete description of the system. However, rarely is an attempt made to solve Equation 1.1.1 but rather simplifications are made from which come the basic Equations of stellar structure. This is generally done by taking 'moments' of the Equations with respect to the various coordinates. For example, noting that the integral of ψ over all velocity space yields the matter density ρ and that no particles can exist with unbounded momentum, averaging Equation 1.1.1 over all velocity space yields

$$\frac{\partial \rho}{\partial \mathrm{t}}+\nabla \cdot(\mathbf{u} \rho)=\overline{\mathrm{S}}, \tag{1.1.2}$$

where u is the average stream velocity of the particles and is defined by

$$\mathbf{u}=\frac{1}{\rho} \int \psi \mathbf{v} d \mathbf{v}. \tag{1.1.3}$$

For systems where mass is neither created nor destroyed $\overline{\mathrm{S}}=0$,

and Equation 1.1.2 is just a statement of the conservation of mass. If one multiplies Equation 1.1.1 by the particle velocities and averages again over all velocity space he will obtain after a great deal of algebra the Euler-Lagrange Equations of hydrodynamic flow

$$\frac{\partial \mathbf{u}}{\partial \mathrm{t}}+(\mathbf{u} \cdot \nabla) \mathbf{u}=-\nabla \Psi-\frac{1}{\rho} \nabla \cdot \mathfrak{P}-\frac{1}{\rho} \int \mathbf{S}(\mathbf{v}-\mathbf{u}) \mathrm{dv}. \tag{1.1.4}$$

Here the forces f have been assumed to be derivable from a potential Ψ. The symbol $\mathfrak{P}$ is known as the pressure tensor and has the form

$$\mathfrak{P}=\int \psi(\mathbf{v}-\mathbf{u})(\mathbf{v}-\mathbf{u}) \mathrm{dv}. \tag{1.1.5}$$

These rather formidable Equations simplify considerably in the case where many collisions randomize the particle motion with respect to the mean stream velocity u .Under these conditions the last term on the right of Equation 1.1.4 vanishes and the pressure tensor becomes diagonal with each element equal. Its divergence then becomes the gradient of the familiar scalar known as the gas pressure P. If we further consider only systems exhibiting no stream motion we arrive at the familiar Equation of hydrostatic equilibrium

$$\nabla \mathrm{P} = -\rho \nabla \Psi . \tag{1.1.6}$$

Multiplying Equation 1.1.1 by v and averaging over v, has essentially turned the Boltzmann transport Equation into an Equation expressing the conservation of momentum. Equation 1.1.6 along with Poisson's Equation for the sources of the potential

$$\nabla^2 \Psi = -4\pi \mathrm{G} \rho , \tag{1.1.7}$$

constitute a complete statement of the conservation of momentum.

Multiplying Equation 1.1.1 by $\mathbf{v} \cdot \mathbf{v}$ or $\mathbf{v}^2$ and averaging over all velocity space will produce an Equation which represents the conservation of energy, which when combined with the ideal gas law is

$$\rho \frac{\mathrm{dE}}{\mathrm{dt}} + \rho \nabla \cdot \mathbf{v} = \rho \varepsilon + \chi - \nabla \cdot \mathbf{F} , \tag{1.1.8}$$

where F is the radiant flux, ε the total rate of energy generation and χ is the energy generated by viscous motions. If one has a machine wherein no mass motions exist and all energy flows by radiation, we have a statement of radiative equilibrium;

$$\nabla \cdot \mathbf{F} = \rho \varepsilon . \tag{1.1.9}$$

For static configurations exhibiting spherical symmetry these conservative laws take their most familiar form:

$$\begin{aligned} \text{Conservation of mass} \quad & \frac{\mathrm{dm(r)}}{\mathrm{dr}} = 4\pi \mathrm{r}^2 \rho . \\ \text{Conservation of momentum} \quad & \frac{\mathrm{dP(r)}}{\mathrm{dr}} = -\frac{\mathrm{Gm(r)}\rho}{\mathrm{r}^2} . \\ \text{Conservation of energy} \quad & \frac{\mathrm{dL(r)}}{\mathrm{dr}} = 4\pi \mathrm{r}^2 \rho \varepsilon , \quad \mathrm{L(r)} = 4\pi \mathrm{r}^2 \mathrm{F} . \end{aligned} \tag{1.1.10}$$

1.2: The Classical Derivation of the Virial Theorem

The virial theorem is often stated in slightly different forms having slightly different interpretations. In general, we shall repeat the version given by Claussius and express the virial theorem as a relation between the average value of the kinetic and potential energies of a system in a steady state or a quasi-steady state. Since the understanding of any theorem is related to its origins, we shall spend some time deriving the virial theorem from first principles. Many derivations of varying degree of completeness exist in the literature. Most texts on stellar or classical dynamics (e.g. Kurth[1]) derive the theorem from the Lagrange identity. Landau and Lifshitz[2] give an eloquent derivation appropriate for the electromagnetic field which we shall consider in more detail in the next section. Chandrasekhar[3] follows closely the approach of Claussius while Goldstein[4] gives a very readable vector derivation firmly rooted in the original approach and it is basically this form we shall develop first. Consider a general system of mass points m_i with position vectors r_i which are subjected to applied forces (including any forces of constraint) f_i. The Newtonian Equations of motions for the system are then

$$\dot{\mathbf{p}}_i = \frac{d(m_i\mathbf{v}_i)}{dt} = \mathbf{f}_i. \tag{1.2.1}$$

Now define

$$G = \sum_i \mathbf{p}_i \cdot \mathbf{r}_i = \sum_i m_i \frac{d\mathbf{r}_i}{dt}\cdot\mathbf{r}_i = \frac{1}{2}\sum_i m_i \frac{d(\mathbf{r}_i\cdot\mathbf{r}_i)}{dt} = \frac{1}{2}\frac{d}{dt}\left(\sum_i m_i r_i^2\right). \tag{1.2.2}$$

The term in the large brackets is the moment of inertia (by definition) about a point and that point is the origin of the coordinate system for the position vectors r_i. Thus, we have

$$G = \frac{1}{2}\frac{dI}{dt}, \tag{1.2.3}$$

where I is the moment of inertia about the *origin* of the coordinate system.

Now consider

$$\frac{dG}{dt} = \sum \dot{\mathbf{r}}_i\cdot\mathbf{p}_i + \sum_i \dot{\mathbf{p}}_i\cdot\mathbf{r}_i, \tag{1.2.4}$$

but

$$\sum_i \dot{\mathbf{r}}_i\cdot\mathbf{p}_i = \sum_i m_i\dot{\mathbf{r}}_i\cdot\dot{\mathbf{r}}_i = \sum_i m_i v_i^2 = 2T, \tag{1.2.5}$$

where T is the total kinetic energy of the system with respect to the origin of the coordinate system. However, since $\dot{\mathbf{p}}_i$ is really the applied force acting on the system (see Equation 1.2.1), we may rewrite Equation 1.2.4 as follows:

$$\frac{dG}{dt} = 2T + \sum_i f_i\cdot r_i. \tag{1.2.6}$$

The last term on the right is known as the Virial of Claussius. Now consider the Virial of Claussius. Let us assume that the forces f_i obey a power law with respect to distance and are derivable from a potential. The total force on the ith particle may be determined by summing all the forces acting on that particle. Thus

$$\mathbf{f}_i = \sum_{j\neq i}\mathbf{F}_{ij}, \tag{1.2.7}$$

where F_{ij} is the force between the ith and jth particle. Now, if the forces obey a power law and are derivable from a potential then,

$$\mathbf{F}_{ij} = \nabla_i m_i \Phi(r_{ij}) = -\nabla_i a_{ij} r_{ij}^n. \tag{1.2.8}$$

The subscript on the ∇-operator implies that the gradient is to be taken in a coordinate system having the ith particle at the origin. Carrying out the operation implied by Equation 1.2.8, we have

$$\mathbf{F}_{ij} = -n a_{ij} r_{ij}^{(n-2)}(\mathbf{r}_i - \mathbf{r}_j). \tag{1.2.9}$$

Now since the force acting on the ith particle due to the jth particle may be paired off with a force exactly equal and oppositely directed, acting on the jth particle due to the ith particle, we can rewrite Equation 1.2.7 as follows:

$$\mathbf{f}_i=\sum_j \mathbf{F}_{ij}=\sum_{j>i} \mathbf{F}_{ij}+\mathbf{F}_{ji}. \tag{1.2.10}$$

Substituting Equation 1.2.10 into the definition of the Virial of Claussius, we have

$$\sum_{\mathrm{i}} \mathbf{f}_{\mathrm{i}} \cdot \mathbf{r}_{\mathrm{i}}=\sum_{\mathrm{i}} \sum_{\mathrm{j}>\mathrm{i}} \mathbf{F}_{\mathrm{ij}} \cdot \mathbf{r}_{\mathrm{i}}+\mathbf{F}_{\mathrm{ji}} \cdot \mathbf{r}_{\mathrm{j}}. \tag{1.2.11}$$

It is important here to notice that the position vector r_i, which is 'dotted' into the force vector, bears the same subscript as the first subscript on the force vector. That is, the position vector is the vector from the origin of the coordinate system to the particle being action upon. Substitution of Equation 1.2.9 into Equation 1.2.11 and then into Equation 1.2.6 yields:

$$\frac{\mathrm{dG}}{\mathrm{dt}}=2 \mathrm{T}-\mathrm{n} \mathcal{U}, \tag{1.2.12}$$

where $\mathcal{U}$ is the total potential energy.[1.1] For the gravitational potential n = -1, and we arrive at a statement of what is known as Lagranges' Identity:

$$\frac{\mathrm{dG}}{\mathrm{dt}}=\frac{1}{2} \frac{\mathrm{d}^2 \mathrm{I}}{\mathrm{dt}^2}=2 \mathrm{T}+\Omega. \tag{1.2.13}$$

To arrive at the usual statement of the virial theorem we must average over an interval of time (T_0). It is in this sense that the virial theorem is sometimes referred to as a statistical theorem. Therefore, integrating Equation 1.2.12, we have

$$\frac{1}{\mathrm{T}_0} \int_0^{\mathrm{T}_0} \frac{\mathrm{dG}}{\mathrm{dt}} \mathrm{dt}=\frac{2}{\mathrm{T}_0} \int_0^{\mathrm{T}_0} \mathrm{T}(\mathrm{t}) \mathrm{dt}-\frac{\mathrm{n}}{\mathrm{T}_0} \int_0^{\mathrm{T}_0} U(\mathrm{t}) \mathrm{dt}. \tag{1.2.14}$$

and, using the definition of average value we obtain:

$$\frac{1}{\mathrm{T}_0}\left[\mathrm{G}\left(\mathrm{T}_0\right)-\mathrm{G}(0)\right]=2 \overline{\mathrm{T}}-\mathrm{n} \overline{\mathcal{U}}. \tag{1.2.15}$$

If the motion of the system over a time T_0 is periodic, then the left-hand side of Equation 1.2.15 will vanish. Indeed, if the motion of the system is bounded [i.e., G(t) < ∞], then we may make the left hand side of Equation 1.2.15 as small as we wish by averaging over a longer time. Thus, if a system is in a steady state the moment of inertia (I) is constant and for systems governed by gravity

$$2 \overline{\mathrm{T}}+\overline{\Omega}=0. \tag{1.2.16}$$

It should be noted that this formulation of the virial theorem involves time averages of indeterminate length. If one is to use the virial theorem to determine whether a system is in accelerative expansion or contraction, then he must be very careful about how he obtains the average value of the kinetic and potential energies.
